



DSA0204-COMPUTER VISION

TECHNICAL PRESENTATION

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TITLE 1:IMAGE PROCESSING VS COMPUTER VISION

INTRODUCTION

Image Processing:

- Enhances or modifies an image.
- Input: Image
- Output: Improved image.

Computer Vision:

- Enables computers to understand image content.
- Input: Image/Video
- Output: Information or decision.



Real-World Applications

Image Processing:

- Photo editing
- Medical image enhancement
- Noise removal

Computer Vision:

- Face recognition
- Self-driving cars
- CCTV surveillance
- Medical diagnosis



CONCLUSION

- Image Processing improves image quality.
- Computer Vision analyzes and understands images.
- Both technologies are widely used in healthcare, security, robotics, and automation.



TITLE 2: COMPARE CCD AND CMOS SENSORS

INTRODUCTION

What is an Image Sensor?

- An image sensor converts light into digital signals.

The two main types are:

- CCD (Charge-Coupled Device)
- CMOS (Complementary Metal-Oxide Semiconductor)



ADVANTAGES

- CCD Advantages
- High-quality images
- Better in low-light conditions
- Low image noise
- CMOS Advantages
- Faster image capture
- Uses less power
- Low cost

Applications CCD:

- Medical imaging, astronomy, scientific cameras
- CMOS: Smartphones, webcams, CCTV cameras, self-driving cars



CONCLUSION

- CCD provides better image quality but is more expensive and consumes more power.
- CMOS is faster, cheaper, and energy- efficient.
- Today, CMOS sensors are widely used in most modern digital cameras and mobile phones